

TARIK AKARLI

Education

Durham University

Sep 2024 – May 2028

Master of Science in Mathematics and Computer Science

Durham, UK

Predicted 1:1. Relevant modules of study and grades can be found [here](#).

Newcastle Sixth Form College

Sep 2022 – Jun 2024

Further Maths (A), Maths (A*), Computer Science (A), Physics(A), Gold award in SMC*

Newcastle upon Tyne, UK

Imperial College London

Jul 2026

Mathematics for Machine Learning Specialization – Coursera

Virtual

- Core Focus: Multivariate calculus optimization, linear algebra for PCA, and algorithmic gradient descent.

Experience and Leadership

Durham University Finance Society

Sep 2025 – Present

Quant Fund member

Durham, UK

- Selected for a competitive programme dedicated to quantitative finance, systematic trading and data-driven research.

Syntropic Labs

Jun 2025 - Present

Quantitative Researcher

London, UK

- Implemented Random Forest and XGBoost models to extract predictive pricing features. Developed and backtested quantitative trading signals using Python, pandas, and SQL.

Solute UK

Jun 2026 - Sep 2026

Digital Engineering Intern

London, UK

- Engineered mission-critical enterprise systems, executing full-lifecycle delivery from initial capacity estimation to production deployment and on-call operational support.

Dialogue Society, Mentor Wise UK, & NEDES

Sep 2024 – Present

Volunteer

Newcastle upon Tyne, UK

- Youth Platform Leader (Dialogue Society): Coordinated community dialogue events and served as Master of Ceremonies for 4 major events, managing programs for audiences of up to 200 people.
- Mentor (Mentor Wise UK): Orchestrated weekly developmental meetings for 7 mentees, cultivating a structured environment to build self-confidence and resolve academic hurdles.
- Mathematics Tutor (NEDES): Managed a specialized classroom environment for cohorts of up to 6 students, driving engagement and cultivating analytical problem-solving frameworks.

Projects

Limit Order Book (LOB) Deep Learning Predictor | *Python, PyTorch, Linux*

Jun 2026

- Architecting a neural network pipeline in PyTorch to forecast short-term mid-price movements utilizing high-frequency tick data.
- Configuring a local, hardware-accelerated Linux compute environment (WSL2, CUDA 12.6) to train deep learning models on a dedicated GPU architecture.

Portfolio Optimizer | *Python, Pandas, NumPy*

Jul 2025

- Implemented mean-variance optimization utilizing NumPy to compute the efficient frontier across a custom 50-asset equity basket, solving for global minimum variance.

High-Frequency Trading Simulator | *Python, NumPy, Pandas*

Apr 2025

- Engineered a vectorized market-making backtester in Pandas, evaluating historical limit order book data to benchmark Sharpe ratio and maximum drawdown metrics.

Monte Carlo Pricing of an Exotic Option | *Python, Pandas, NumPy*

Jan 2025

- Built a Streamlit app to Monte Carlo-price discrete arithmetic Asian options under GBM, with dividends and call/put support.
- Added variance reduction (antithetic, moment matching, control variate). Produced clear diagnostics and exports: convergence/distribution plots, sample paths, CSV

Any projects and their applications and any academic info can be found on my portfolio [here](#).

Technical Skills

Languages: Python, C++, R, SQL, JavaScript

Libraries and Frameworks: PyTorch, Pandas, NumPy, Matplotlib

Technologies and Tools: Linux, Git, LaTeX, CUDA

Spoken Languages: English (Fluent), Turkish (Fluent)